BTS-I&II(R/S)-(04/09)-21-0807 Reg. No. [ | | | | | | | ]

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# B.Tech. Degree I & II Semester Regular/Supplementary Examination April/September 2021

**ME/SE 19-200-0103A ENGINEERING GRAPHICS**

*(2019 Scheme)*

*(Regular)*

Time: 3 Hours Maximum Marks: 60

## PART A

$(1 \times 12 = 12)$

I. (a) Construct a scale of 1:4, to show centimetres and long enough to measure up to 5 decimetres. Mark on the scale a measurement of 3.7 decimeters. (6)

(b) Point P is 80 mm from point O. It starts moving towards O and reaches it in two revolutions around it. Draw locus of point P. (6)

**OR**

II. (a) Draw a Vernier scale of R.F. = 1/25 to read up to 4 meters. On it show lengths 2.39 m and 0.91 m. (6)

(b) Draw locus of a point, 5 mm away from the periphery of a circle which rolls on straight line path. Take circle diameter as 50 mm. (6)

## PART B

$(4 \times 15 = 60)$

III. (a) A line 80 mm long makes an angle of 35° with the H.P. and 45° with the V.P. Its mid-point is 30 mm above the H.P and 25 mm in front of the VP. Draw the projections of the line. (10)

(b) A circular lamina of 50 mm diameter is resting on HP. The surface of the lamina is inclined 40° to HP. Draw the projections of the circular lamina. (5)

**OR**

IV. (a) The dimensions of a room are 100 m × 80 m × 60 m. Find the length of the largest diagonal inside the room using graphical method. (6)

(b) A pentagonal lamina of 35 mm edge length is resting on one of its edge on HP. The surface of the lamina is inclined 35° to HP. Draw the projections of the plane. (9)

V. A frustum of a cone, diameter of base 60 mm, diameter of top surface 30 mm and axis 45 mm long is lying on HP on one of its generators. The plane containing the axis and the generator makes an angle of 45° to VP. Draw its front and top views. (15)

**OR**

VI. A pentagonal prism of 40 mm base edge and 70 mm height is resting on HP on its base with one base edge perpendicular to VP. A cutting plane inclined 30° to HP and passing through the centre of the axis cuts the solid. Draw the front view, sectional top view and true shape of the section. (15)



(P.T.O.)

VII. A waste basket is in the form of a frustum of a cone. It is having bottom diameter 60 mm and top diameter 85 mm. The height of the basket is 50 mm. Draw the development of the basket. (Assume the position of a waste basket). (15)

**OR**

VIII. A square prism, edge of base 30 mm and height 60 mm, resting on its base on HP, is completely penetrated by another square prism of 20 mm base edge such that the axis of the penetrating prism is perpendicular to and 6 mm in front of the axis of the vertical prism. The rectangular faces of the two prisms are equally inclined to the VP. Draw the projections of the solids showing line of intersection. Length of both prisms is 100 mm. (15)

IX. A hemisphere of 44 mm diameter is nailed on the top face of a frustum of a hexagonal pyramid with flat surface up. Sides of top and bottom faces of the pyramid are 15 mm and 30 mm respectively and its height is 50 mm. The axes of both the solids coincide. Draw the isometric projection. (15)

**OR**

X. Draw the perspective view of a rectangular prism of 100 mm × 50 mm × 40 mm size lying on its 100 mm × 50 mm rectangular face on the ground plane, with a vertical edge touching the picture plane and the end faces inclined at 45° with the picture plane. The station point is 120 mm in front of the picture plane, 80 mm above the ground plane and lies in a central plane which is passing through the centre of the prism. (15)

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